

PolicyPulse AI

Team Handover & Briefing Document

For business development, software engineering, product, and leadership

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Repository: <https://github.com/ru-ri-p/policypulse-ai>

AI policy monitoring for compliance teams in the UAE/GCC and beyond.

1. Executive Summary

PolicyPulse AI is an AI policy monitoring and compliance intelligence platform built for organisations in Dubai/GCC that must track local regulations (UAE, DIFC, ADGM, Saudi) and extraterritorial exposure (EU AI Act, US federal AI policy, UK ICO). It automatically ingests official policy sources, detects changes, classifies risk, tags by zone and sector, maps playbook impact, and can produce LLM compliance digests and weekly PDF briefs.

Positioning: MENA-first AI policy monitor with zone/playbook layers and extraterritorial exposure packs — not a generic global feed, not internal-only GCC governance, not search-only legal tech.

Status: Working prototype, demo-ready locally. Layer 2 (playbooks, zones, weekly PDF), publication dates, and Phase 9 (MENA law spiders + Celery Beat) are implemented locally but not yet pushed to GitHub. No production deploy URL yet.

2. What the Product Does

Capability	Description
Ingestion	Scrapy spiders pull policy text from official portals into PostgreSQL
Change tracking	Archives prior versions; flags % change and major updates (>20%)
Classification	Zero-shot ML: jurisdiction, doc type, risk level
Semantic search	pgvector + sentence-transformers (all-MiniLM-L6-v2)
Compliance digests	GPT-4o-mini: what changed, who affected, what to do, urgency
Personalised feed	JWT users + profile → relevance scoring
Alerts	Nightly email for high-relevance + urgent changes (SMTP required)
Zones & playbooks	DIFC/ADGM/MAINLAND tags + rule-based impact levels
Exposure packs	Client profiles: dubai_hq, mena_only, eu_exposure_only
Weekly PDF brief	Branded ReportLab report for investors/clients
Dashboard	Streamlit: feed, search, changes, stats, weekly brief tab
REST API	FastAPI v0.4.0 — documents, search, auth, playbooks, stats

3. Target Market & GTM

Segment	Value proposition
DIFC/ADGM fintech	Zone playbooks + EU/US extraterritorial packs
Dubai HQ multinationals	One feed: UAE + EU AI Act + US federal exposure
Law firms / consultancies	White-label policy intelligence (B2B2B API)
AI governance leads	Change alerts, digests, version audit trail

Default demo pack: dubai_hq (GCC + EU + US + UK — not EU-only).

4. Technical Architecture

Sources (Scrapy) → Postgres pipeline → Change detector → Celery + Redis (spiders, ML, digests, alerts) → PostgreSQL 16 + pgvector → FastAPI → Streamlit dashboard

Component	Technology
Language	Python 3.13
Ingestion	Scrapy, custom pipelines
Database	PostgreSQL 16, pgvector
Queue	Redis + Celery Beat
API	FastAPI, JWT, Pydantic
UI	Streamlit, Plotly
ML	sentence-transformers, BART-MNLI (zero-shot)
LLM	OpenAI GPT-4o-mini (optional digests)
PDF	ReportLab
Tests	pytest (51 tests)
Deploy	Docker Compose (local); production TBD

5. Everything Built (Phases 1–9 + Layer 2)

Phase 1–3 — Foundation

- PostgreSQL schema, Scrapy pipeline, change detection
- Celery + Redis scheduling
- FastAPI: JWT, documents, search, stats, alerts
- Spiders: EU AI Act, US Federal Register, NIST, UK ICO

Phase 4 — LLM digests

- Structured digest columns (what/who/todo/urgency/deadline)
- digests/run_digests.py + Celery batch task

Phase 5 — Relevance & alerts

- User profiles, personalised /user/feed
- Nightly relevance email alerts

Phase 6 — Dashboard & Docker

- Streamlit dashboard
- Docker Compose full stack

Phase 7 — Polish

- 51 pytest tests
- OECD + UK AISI spiders

- Case study + launch docs

Phase 8 — MENA coverage

- region column, MENA spiders (UAE, DIFC, ADGM, SDAIA, Digital Dubai)
- Classifier MENA labels, API region filter
- Dashboard MENA default (later changed to All)

Layer 2 — Zones, playbooks, weekly PDF

- zones, sectors, playbook_impacts on documents
- YAML playbooks + exposure packs (dubai_hq includes EU+US)
- API /playbooks, zone/sector filters
- Weekly PDF generator (ReportLab)
- Dashboard zone filter, playbook impacts, weekly brief tab

Layer 2.5 — Dates & UX

- first_scraped_at, publication date extraction
- Dashboard: Published vs Ingested, update status badges
- Keyword filter, zone guide in sidebar

Phase 9 — MENA law-focused ingestion

- New spiders: uae_legislation, dfsa_rulebook
- Refocused DIFC, ADGM, SDAIA, Digital Dubai (law URLs only)
- Celery Beat: MENA daily 06:00–07:50 UTC + retag
- scripts/run_mena_spiders.sh manual runner

6. Current Data Snapshot

Live from local database at generation time.

Metric	Count
Total documents	1173
With publication date	1070
MENA (UAE + SA)	22
With compliance digest	0
With embeddings (searchable)	40

By jurisdiction

Jurisdiction	Documents
US	1114
UAE	22
EU	20
UK	17

By region

Region	Documents
US	1114
MENA	22
EU	20
UK	17

MENA sources (documents saved)

Source	Count
UAE AI Office	10
Digital Dubai	10
DIFC Data & AI Regulation	2

MENA zone tags

Zone	Count
MAINLAND	20
DIFC	2

7. Spider Inventory & Celery Schedule

Spider	Focus	Beat (UTC)
uae_legislation	UAE federal law portal	06:00
dfsa_rulebook	DFSA DIFC rulebook	06:15
difc_laws	DIFC laws & regulations	06:30
adgm_fsra	ADGM legal framework / FSRA	06:45
sdaia_saudi	Saudi PDPL / SDAIA	07:00
uae_ai_office	UAE government AI policy	07:15
digital_dubai	AI principles & governance	07:30
—	retag_all_documents	07:50
eu_ai_act	EU AI Act	09:00
federal_register	US Federal Register	09:30
nist_airc	NIST AI Resource Center	10:00
ico_uk	UK ICO AI guidance	10:30
oecd_policy, uk_aisi	OECD / UK AISI	Manual

8. API Surface (v0.4.0)

Endpoint	Purpose
GET /documents/	List/filter: region, jurisdiction, zone, sector, risk, keyword
GET /documents/{id}	Single doc: dates, zones, impacts, digest
GET /search/	Semantic search (requires embeddings)
GET /stats/	Totals by region, jurisdiction, source
GET /playbooks/	Regulatory playbooks list
GET /playbooks/exposure-packs/{id}	Client exposure pack detail
GET /playbooks/impacts/{doc_id}	Playbook impacts per document
POST /auth/login	JWT authentication
GET /user/feed	Personalised relevance feed
GET /health	Health check

9. Exposure Packs & Playbooks

Pack ID	Use case
dubai_hq	Dubai HQ: GCC + EU + US + UK (recommended demo default)
mena_only	GCC-only footprint
eu_exposure_only	EU AI Act / GDPR extraterritorial focus

Playbooks (rule-based YAML)

- uae_mainland_general — UAE federal PDPL & AI Office
- difc_financial_services — DIFC / DFSA
- adgm_financial_services — ADGM / FSRA
- cbuae_banking_ai — UAE supervised financial institutions
- digital_dubai_tech — Dubai AI ethics & smart city
- sdaia_saudi — KSA PDPL & SDAIA
- eu_ai_act_exposure — EU extraterritorial
- us_federal_ai_exposure — US partner/parent exposure
- uk_ico_ai — UK ICO & data protection

10. What Works Well Today (Demo-Ready)

- Policy feed with region, zone, exposure pack, risk, keyword filters
- US Federal Register with real publication dates and change flags
- Playbook impact badges (rule-based, no OpenAI required)
- Change tracking: new / updated / major revision + version archive
- Weekly PDF generation for dubai_hq exposure pack
- 51 automated tests passing
- Docker Compose for local full stack
- Celery Beat schedule defined for daily ingestion

11. Downfalls, Gaps & Risks

Data & coverage

Issue	Impact
~95% US corpus	MENA demos look thin vs positioning
MENA = mostly marketing pages	Credibility gap for legal/compliance buyers
Law spiders yield few new docs	DFSA/ADGM/SDAIA/uaelegislation not saving much yet
No PDF pipeline	Binding instruments often PDF-only
MENA publication dates sparse	Sources omit dates in HTML
Only ~40 docs embedded	Semantic search mostly empty
0 digests generated	Needs OPENAI_API_KEY + batch run

Product & UX

Issue	Impact
Streamlit ≠ production SaaS	No multi-tenant UI, SSO, billing
Search ignores sidebar filters	User confusion
No Arabic	GCC enterprise expectation
No acknowledge / audit trail	Enterprise workflow missing
PDF via shell script only	Not one-click in dashboard

Engineering & ops

Issue	Impact
Layer 2 + Phase 9 not on GitHub	Team cannot reproduce latest easily
No production deploy URL	No shareable live demo
Celery worker must be running	Beat schedule inactive otherwise
SDAIA/DFSA Cloudflare	Bot blocking in production
ML slow on CPU	Batch only, not real-time
No observability	No Sentry/Prometheus/spider failure alerts

12. Recommended Roadmap

Before Git push & pilot

- Commit + push Layer 2, dates, Phase 9
- Deploy API + dashboard (Railway/Render)
- Run OpenAI digests on top 50 high-risk / dubai_hq docs
- Embed full corpus for search demo
- Record 5-min Loom demo + README screenshots

MENA credibility (highest product priority)

- PDF ingestion pipeline (DFSA, ADGM, federal decrees)
- CBUAE spider for banking AI guidance
- Fix law spider selectors after live HTML inspection
- Manual seed of 10–20 canonical instruments (PDPL, DIFC DP Law)

Layer 3 — sellable product

- Acknowledge workflow (who reviewed what, audit export)
- Email/Slack alerts on high playbook impact + major changes
- B2B2B API keys per tenant + webhooks
- Bilingual digests (EN + AR)
- One-click weekly PDF from dashboard

Engineering hardening

- Spider success/failure ops dashboard
- Production UI (Next.js) when ready
- Fine-tuned MENA classifier
- Rate limiting, SSO, RBAC

13. Briefing Notes by Role

Business development

- Pitch: only AI policy monitor for Dubai HQ with DIFC/ADGM zones + EU+US packs
- Demo: Region All → dubai_hq → US doc with date → playbook impacts → weekly PDF
- Pilot target: DIFC/ADGM fintech compliance lead or consultancy
- Do not oversell: MENA law library thin; US change monitoring strongest
- Pricing: per-seat SaaS, per-exposure-pack API, or white-label

Software engineers

- Entry: ingestion/spiders/, api/routers/, dashboard/app.py, infra/celery_app.py
- Run: ./scripts/run_api.sh, run_dashboard.sh, run_tests.sh
- MENA: ./scripts/run_mena_spiders.sh
- Migrations: ingestion/migrations/002–010
- Large uncommitted diff vs origin/main — sync before collaborating

Product / compliance advisors

- Playbooks are YAML — editable without code (ingestion/playbooks/)
- dubai_hq intentionally includes EU + US, not EU-only
- Publication date ≠ ingest date on dashboard
- Digests optional — core monitoring works without OpenAI

14. Competitive Landscape

Competitor	Strength	PolicyPulse edge / gap
Verifli	Global AI policy monitor	No GCC zone/playbook layer — we differentiate
GCC LexAI	Legal search	We do change monitoring + playbooks
ALFABEN / OSSUS	GCC governance, Arabic	We are external policy + extraterritorial
Generic RegTech	Enterprise sales	We are AI-policy-specific, MENA-first

15. Commands Cheat Sheet

```
./scripts/run_api.sh # API :8000  
./scripts/run_dashboard.sh # Dashboard :8501  
./scripts/run_mena_spiders.sh # MENA law crawl + retag
```

```
./scripts/run_tests.sh # pytest (51 tests)

./scripts/generate_weekly_brief.sh --exposure-pack dubai_hq --days 7

python3 -m ingestion.retag_documents

python3 -m ingestion.backfill_published_at

python3 -m digests.run_digests --limit 50 # needs OPENAI_API_KEY

docker compose up --build
```

One-line status for leadership: PolicyPulse AI is a working compliance intelligence prototype with strong US ingestion and differentiated MENA architecture (zones, playbooks, exposure packs), but MENA legal depth, production deployment, digests, and enterprise workflow features are needed before a paid pilot.

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